

# Network Canvas Architect: From concepts to practices

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# Why Networks?

- Link micro-level behaviours to emergent macro-level outcomes (“Coleman’s boat”).
- Consider individuals in their social context.
- Understand emergent phenomena, such as swarming, imitation, diffusion.
- Represent social life with high fidelity, empirically-grounded relationships.

# Canonical Example 1

## **The strength of weak ties**

- People tend to want their close friends to know each other.
- As they 'close these triads' they create clustering.
- Information circulates faster within clusters.
- So to find novel information, such as a job opening, we can look to the unclosed triads of weak ties rather than the highly clustered strong ties.



# Canonical Example 2

## **The six degrees of separation**

- Everyone in the world is connected to at least someone else by virtual of birth.
- On average, how many hops would it take to get from one to another.
- Milgram asked people to send a package from Nebraska to Boston.
- The median chain of referrals was six. Hence the six degrees of separation.



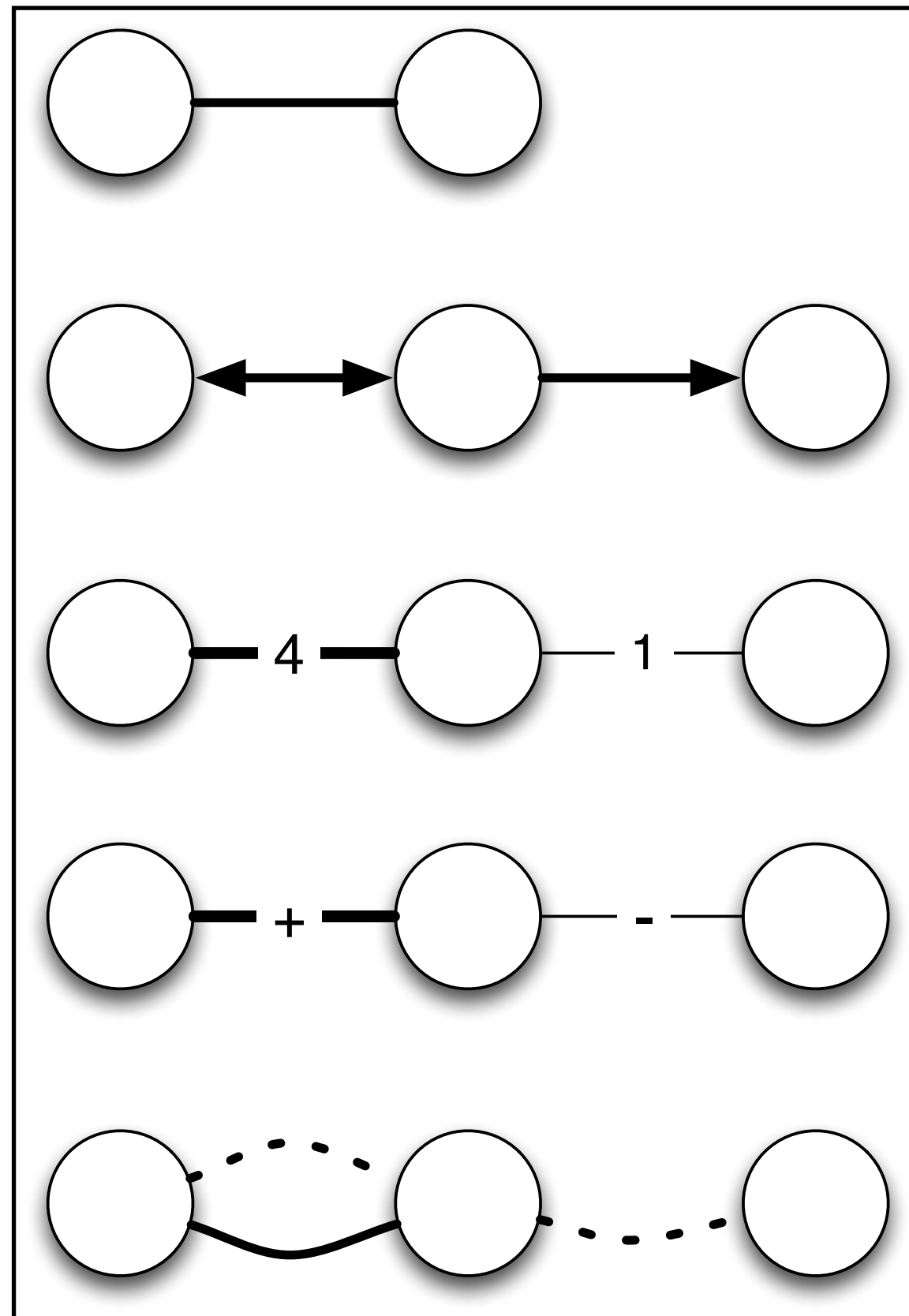
# Networks as a Social Construction

- Networks constitute a method and a theory
- The world is not made of networks, but regularities that can often be described as such.
- Networks have persisted
  - Because of their analytic power
  - Because of their synthetic utility

# Basic Terms

- ***Node (or vertex)***: Basic unit of analysis. Usually of similar type. Can be any boundable object: people, countries, organizations, departments, email messages.
- ***Tie (or edge / arc)***: relation connecting two nodes.
- ***Mode***: The type of relations. One-mode mean all nodes type. Two-mode means two kinds of nodes (such as authors and journals )
- In mathematical notation,  $G$  is a graph.  $G$  is also a set that contains  $V$  vertices and  $E$  edges. It can be expressed as
  - $G = (V, E)$

# What's in a relation?



**Undirected:** “Edge” - e.g., Facebook friends.  $V_{ij} = V_{ji}$

**Directed:** “Arc” - e.g., Gives advice.  $V_{ij} \neq V_{ji}$

**Weighted:** Numerical value of relation, e.g., number of messages sent. Higher value = “stronger” tie between people.

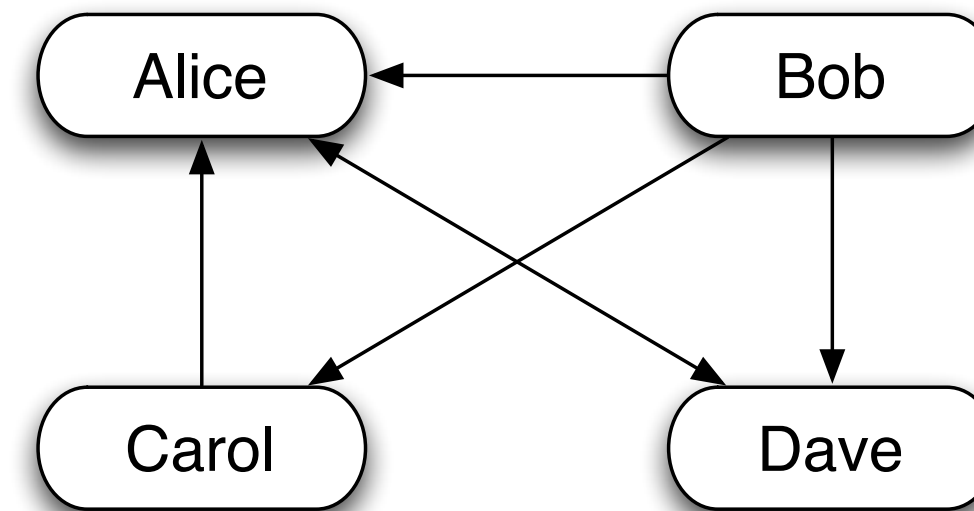
**Valued:** Positive or negative, e.g., Emotional energy; positive / negative reviews on eBay.

**Multiplex:** Multiple relation types between people.



# Representing Network Data

|       | Alice | Bob | Carol | Dave |
|-------|-------|-----|-------|------|
| Alice | -     | 0   | 0     | 1    |
| Bob   | 1     | -   | 1     | 1    |
| Carol | 1     | 0   | -     | 0    |
| Dave  | 1     | 0   | 0     | -    |



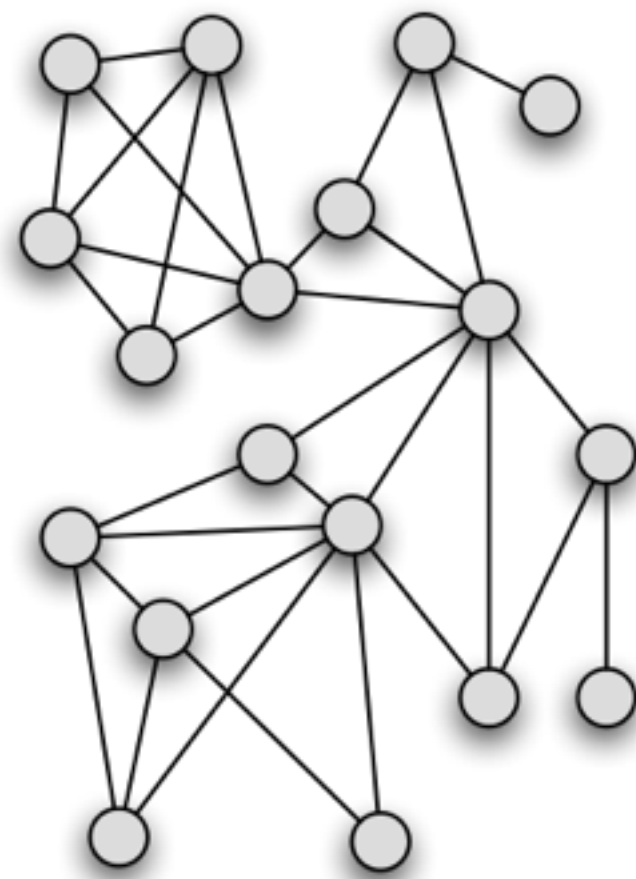
| From  | To    |
|-------|-------|
| Alice | Dave  |
| Bob   | Alice |
| Bob   | Carol |
| Bob   | Dave  |
| Carol | Alice |
| Dave  | Alice |
| Dave  | Bob   |
| Dave  | Carol |

## Network links can be represented as:

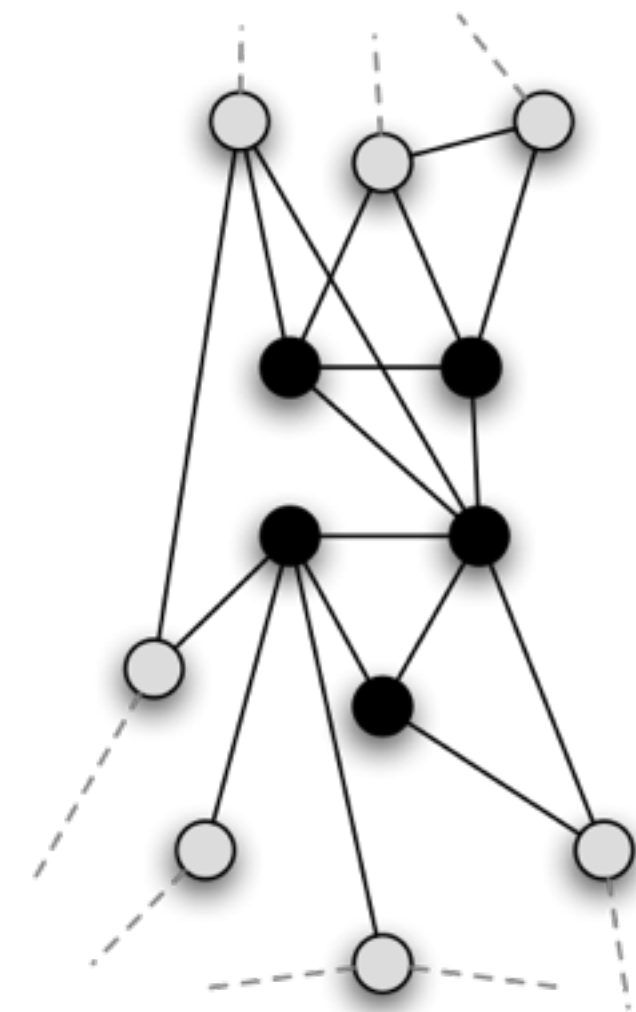
- Adjacency matrices: *from* in rows and *to* in columns,
- Sociograms: relations represented visually,
- Edge lists: *from* in column 1 and *to* in column 2, plus attributes in additional columns.

# Network collection types

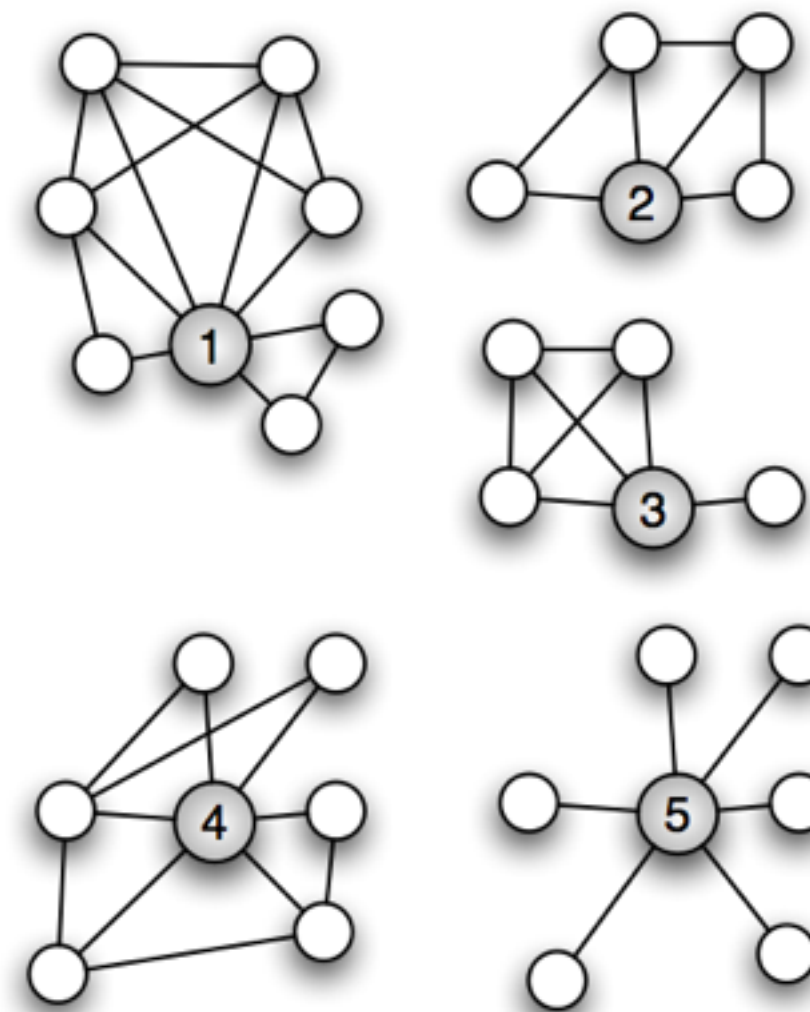
Whole Network



Partial Network



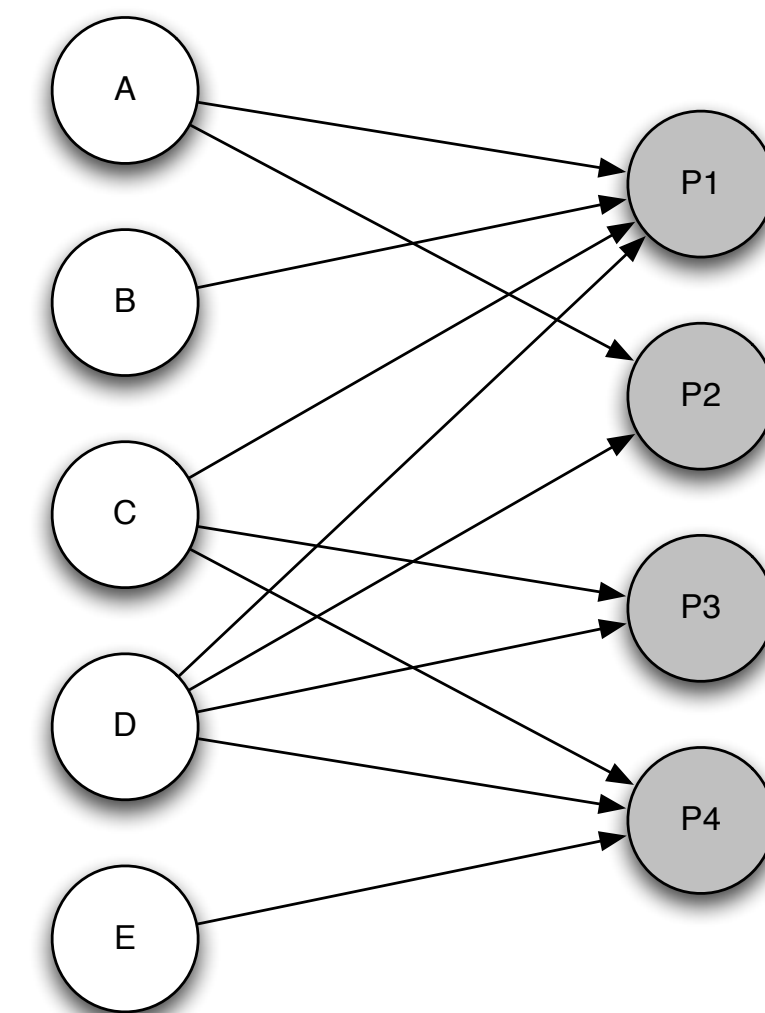
Five Ego Networks



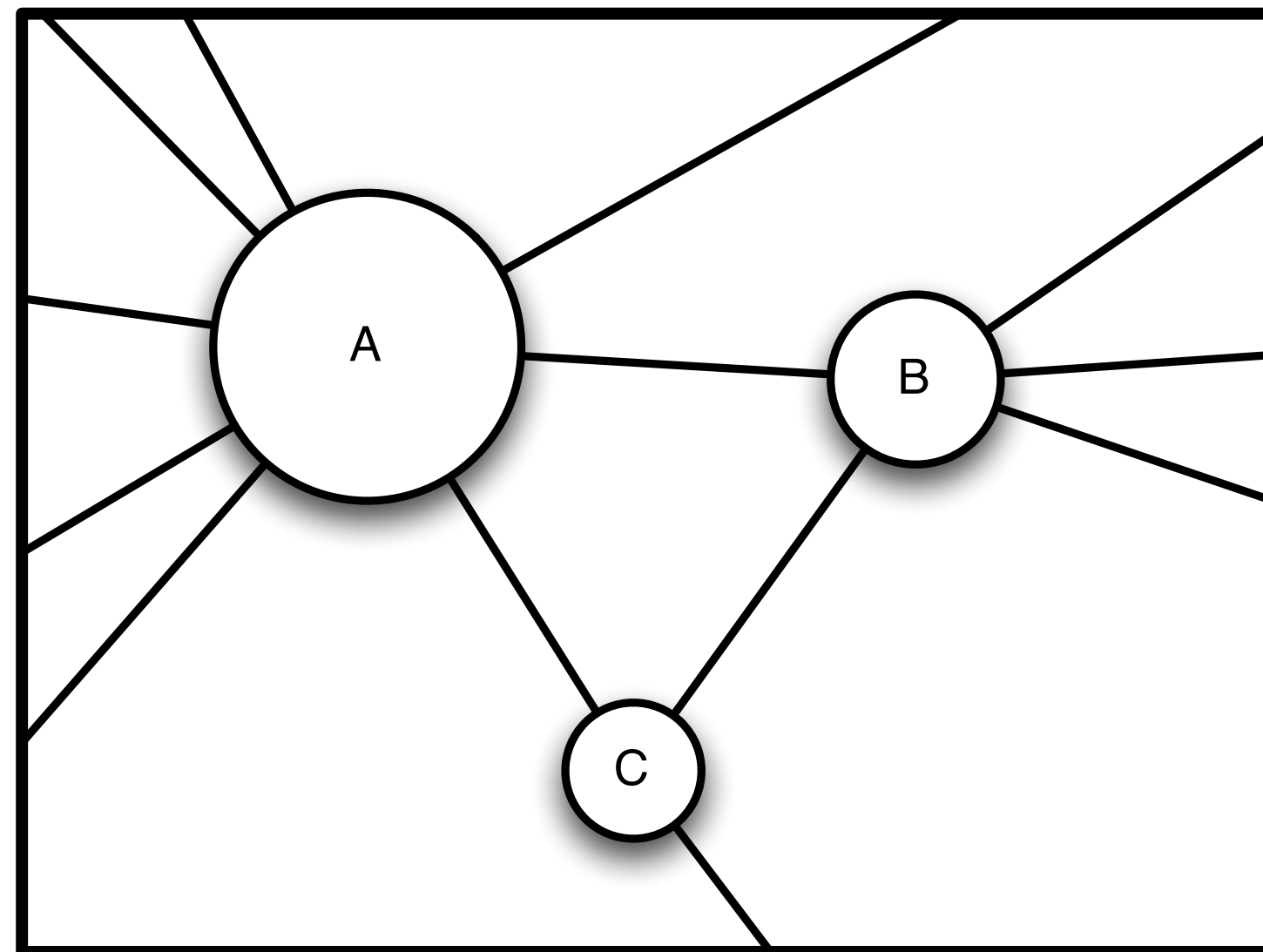
Two-mode networks

Actors (Nodes)

Events (Parties)



# Units of analysis: Position



## Position:

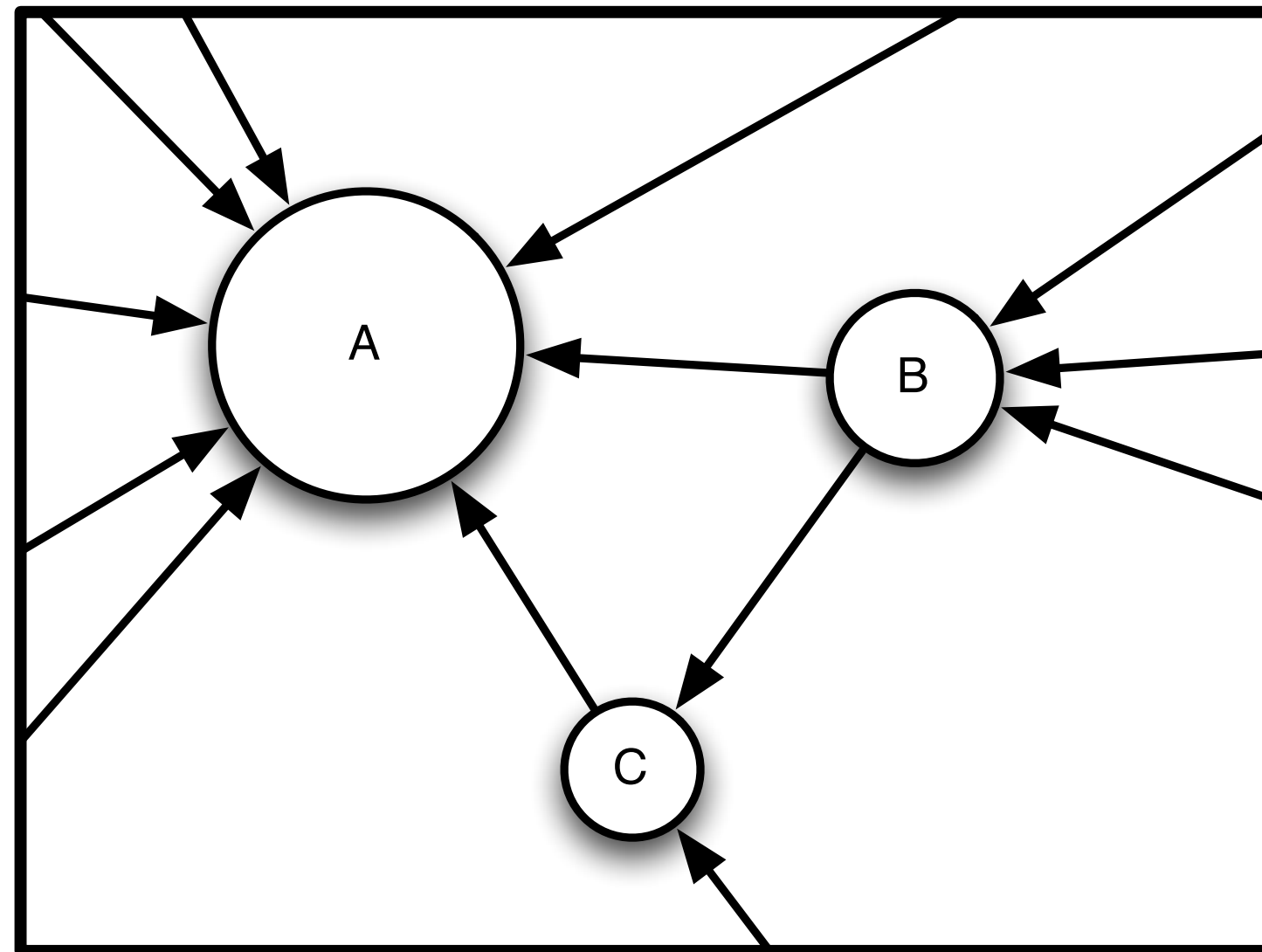
Looking for a node's position is one of the first and most sensible tasks for the analyst.

Nodes with more links are considered more *central*.

*Degree centrality* is a score for more links, where a link is considered a 'degree'.



# Units of analysis: Position

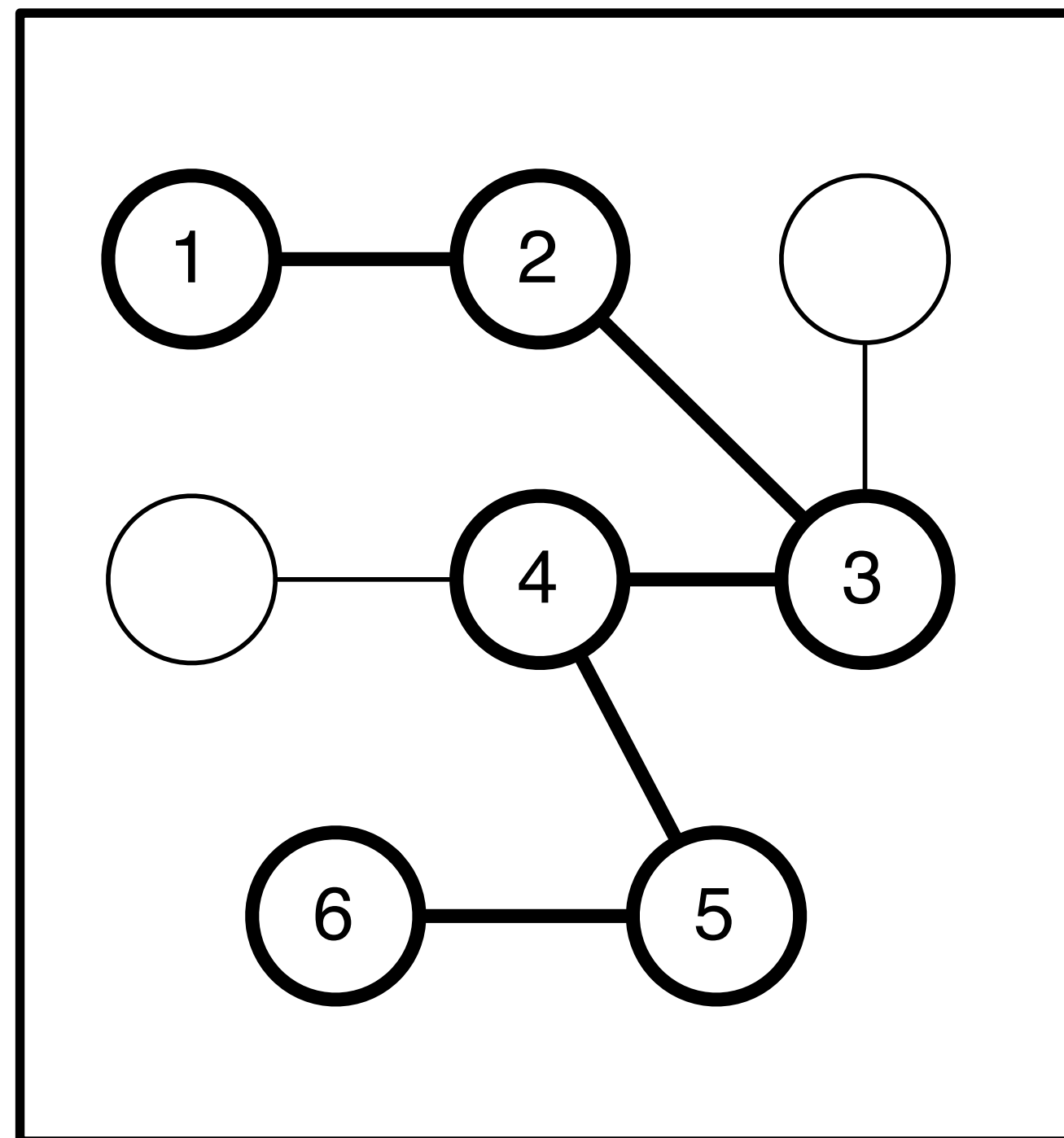


## In- and out-degree:

Some links are directed.

When links represent a message, nodes with a higher *in-degree* are more popular. Nodes with more *out-degree* are more authoritative.

# Definition: Geodesic



## **A geodesic:**

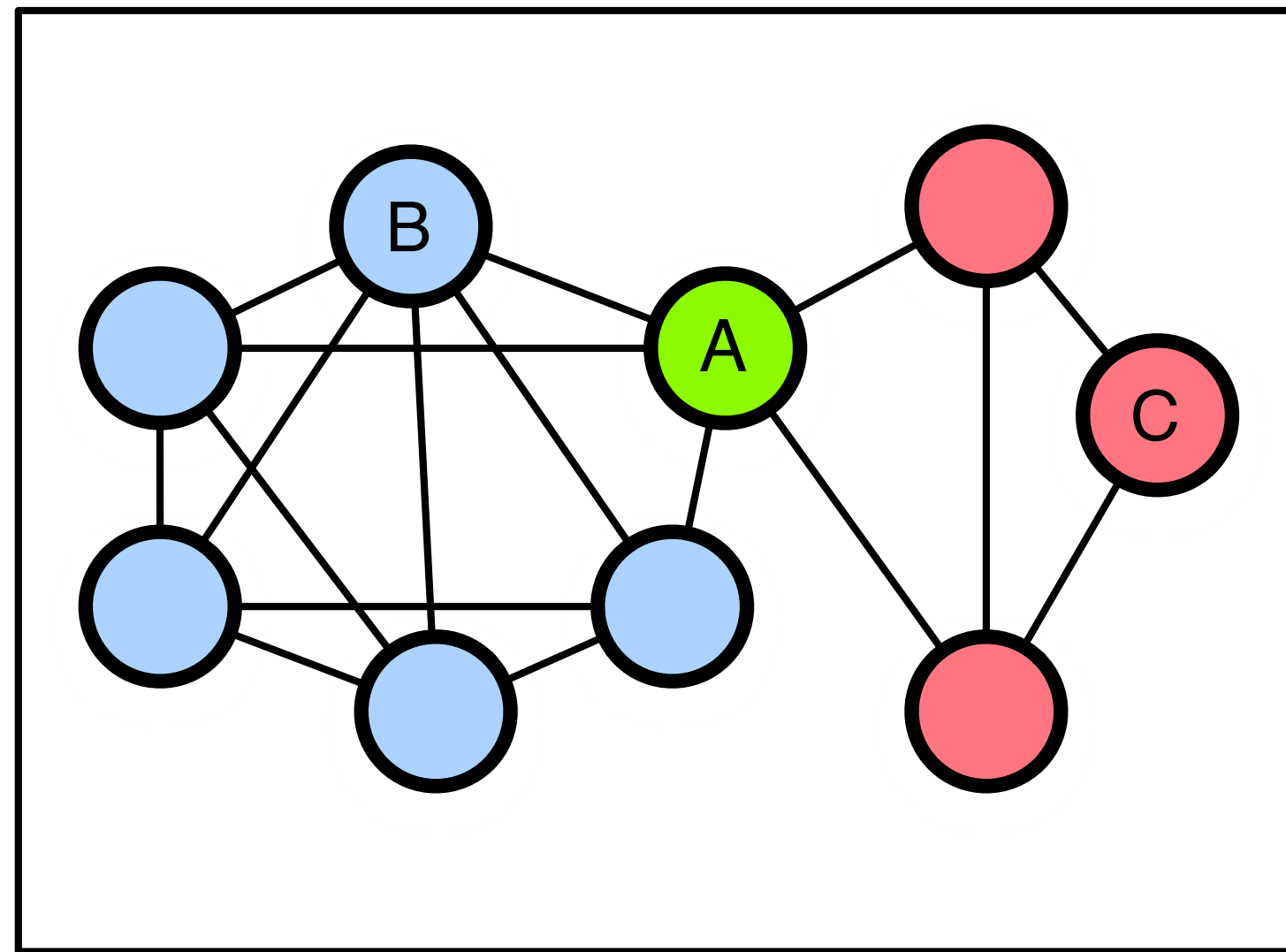
Is the shortest path between two nodes.

## **A diameter:**

Is the longest of these short paths in the network.



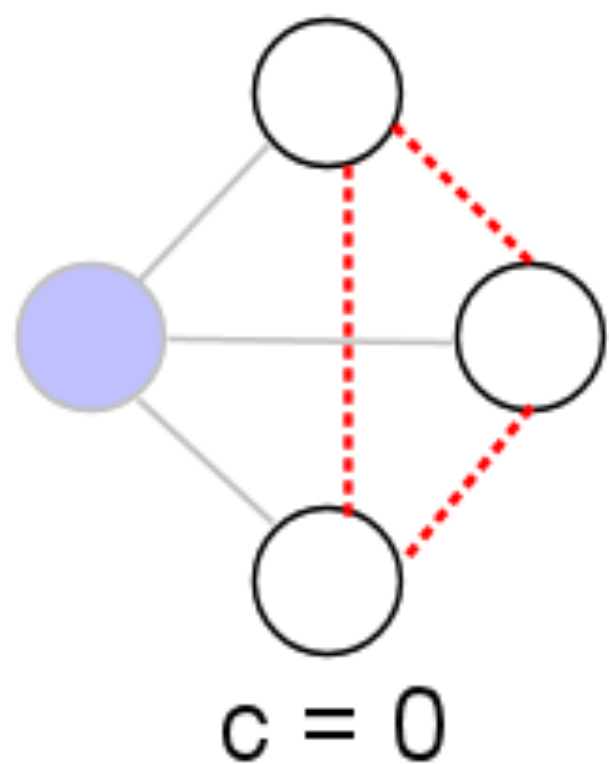
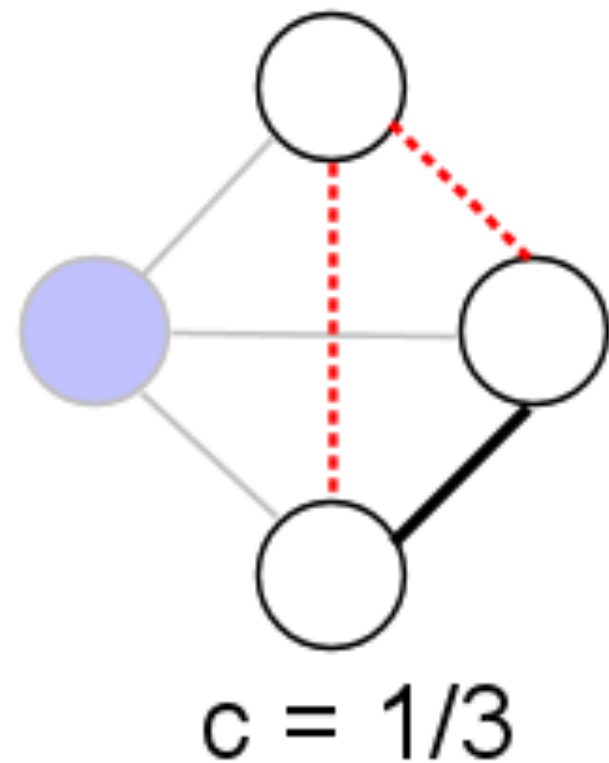
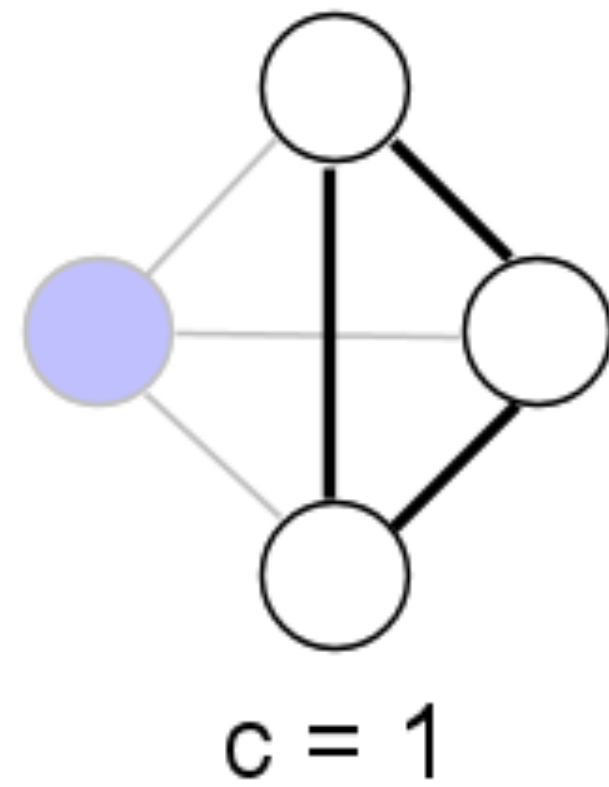
# Units of analysis: Position



## **Betweenness:**

Being more popular is not always the best measure of centrality. What if you are the only link between two sources of information?

*Betweenness centrality* measures where individuals lie on shortest paths.



# Clustering Coefficient

Percentage of all triads that are realised as triples.  
*View the diagram from the point of view of the purple node.*

## **Local clustering coefficient:**

Percent of all nodes around a given node. Acts as a measure of local density.

## **Global clustering coefficient:**

Average of all LCC scores across the entire network.

## **Transitivity:**

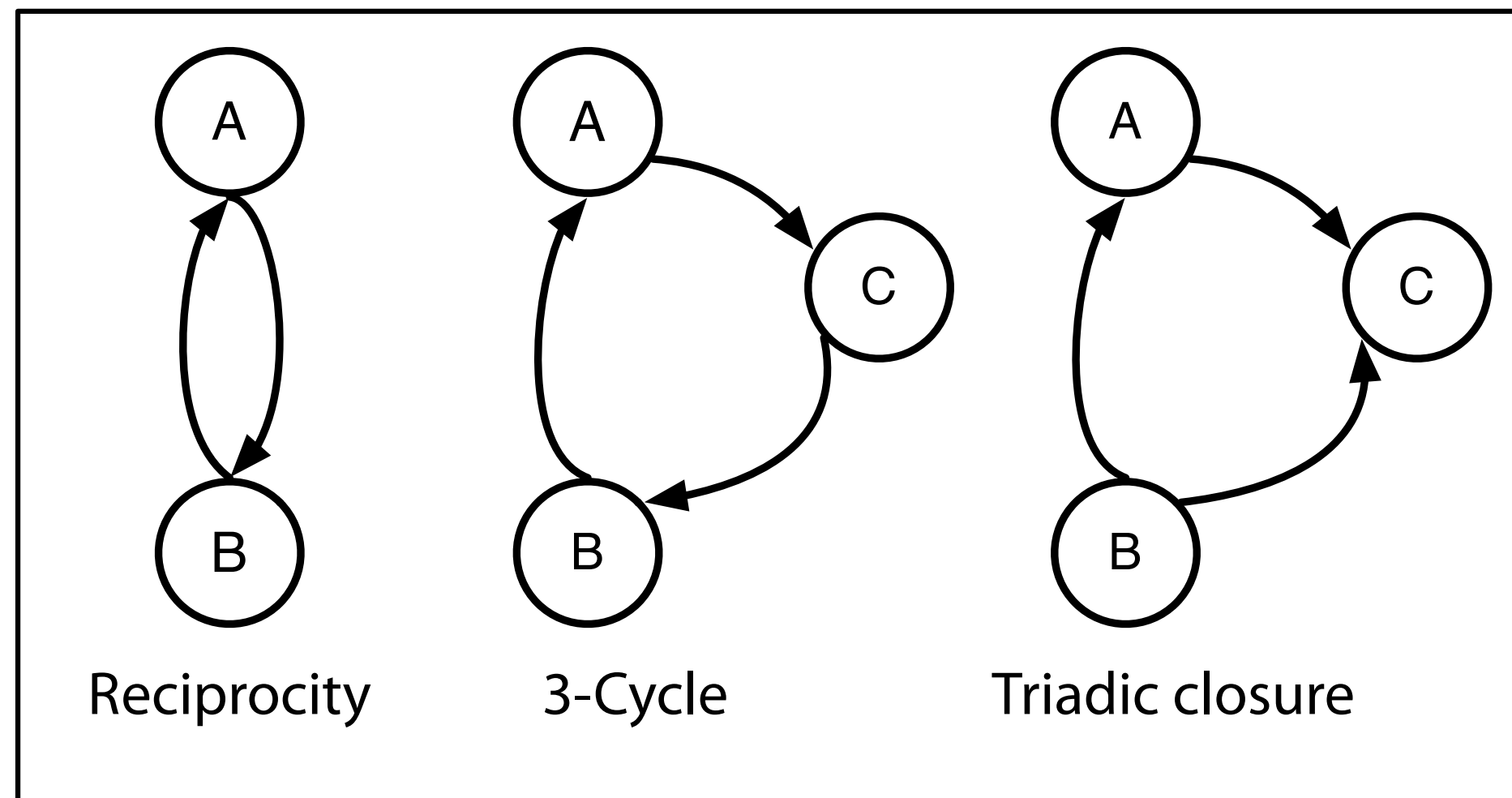
(Number of triads \* 3) / number of triples. This is like the LCC except it's calculated for all the network rather than for all the egos and averaged.

# How to make a network?

- While some networks can be derived from empirical data or manifested in socio-technical systems, others are synthetic graph objects. Some examples:
  - A star graph is a graph with one centre node and  $n$  nodes that connect to the centre. (`Star` in igraph)
  - A Bernoulli graph / Erdos-Renyi graph is a graph where edges exist with probability  $p$  given all possible edges. (`Erdos_Renyi` in igraph)
  - A Molloy graph is a randomly generated graph where the number of nodes are fixed as is the *degree distribution*. (`rewire` in igraph)



# Units of analysis: Micro



## Configurations:

Certain arrangements of nodes are important, such as reciprocal edges, 3-cycles and triadic closure.

*Reciprocity* is extremely prevalent in networks.

*3-cycles* imply egalitarianism.

*Triadic closure* is common and implies hierarchy.

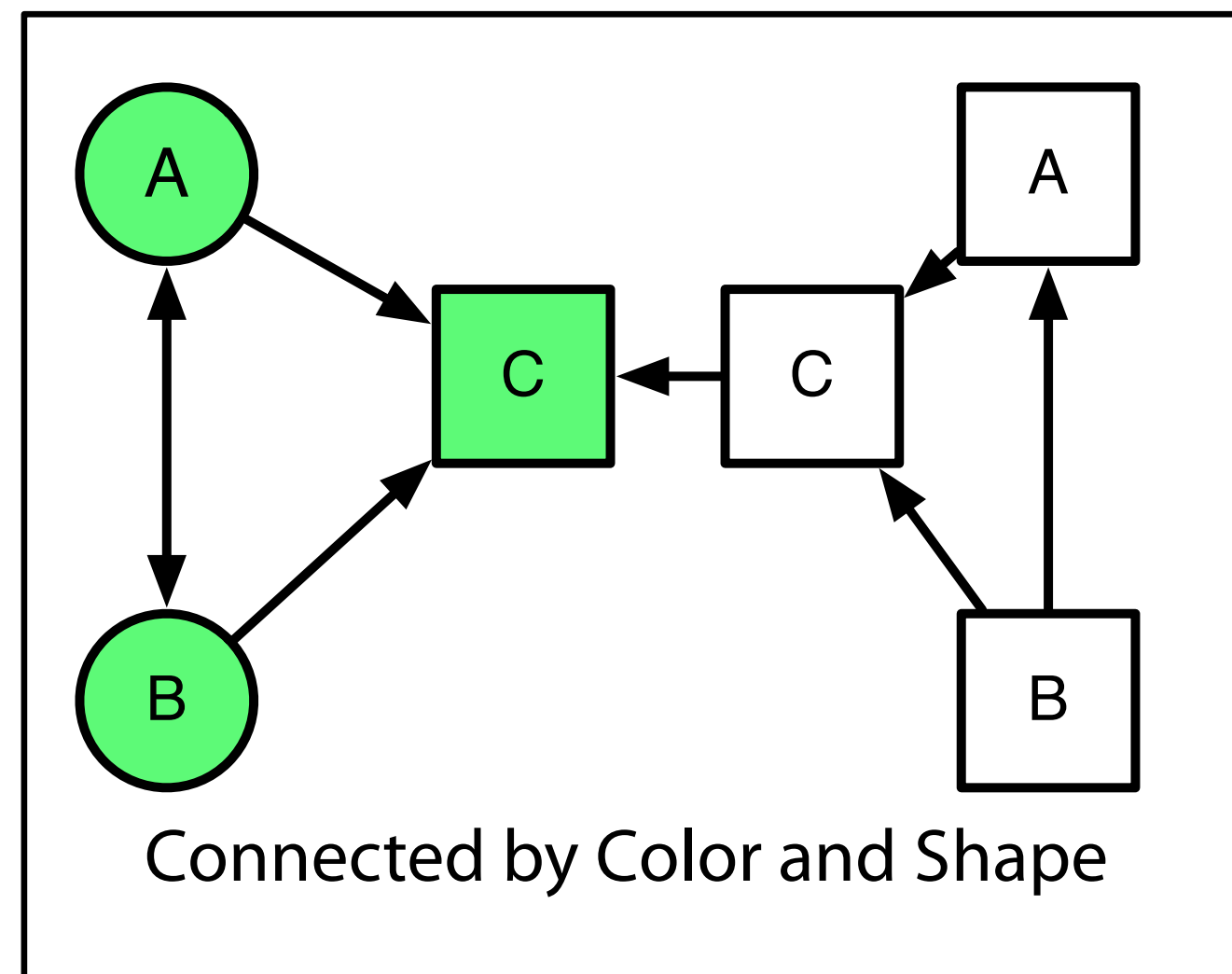
# Homophily



Birds of a feather

Flock together

# Units of analysis: Micro



## Homophily:

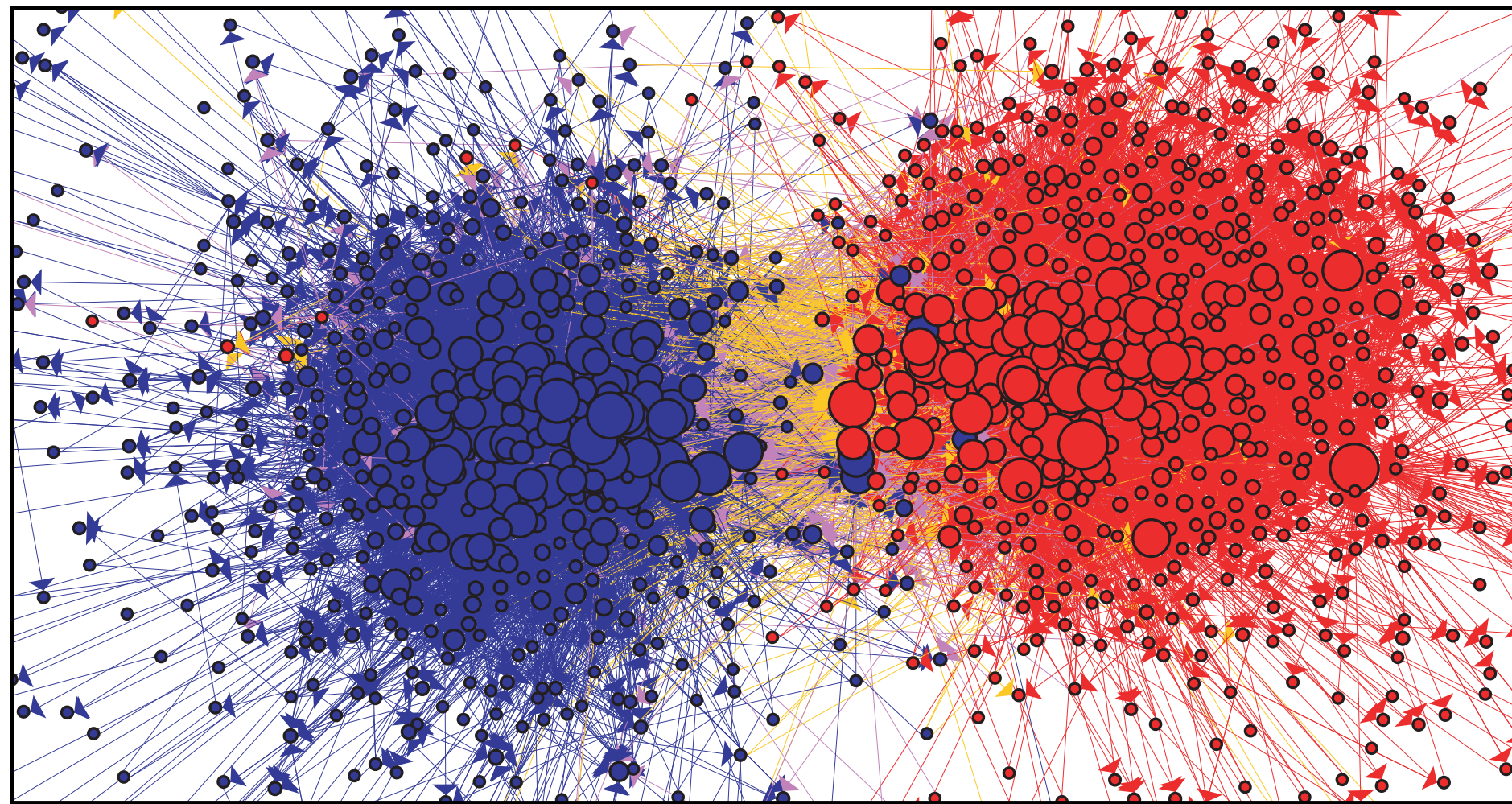
Individuals of like type are particularly prone to linking to one another.

The trick is not finding homophily, but the right kind of homophily.

*Baseline homophily* is how much you would expect by chance. In-breeding homophily is homophily over and above this value.



# Units of analysis: Meso

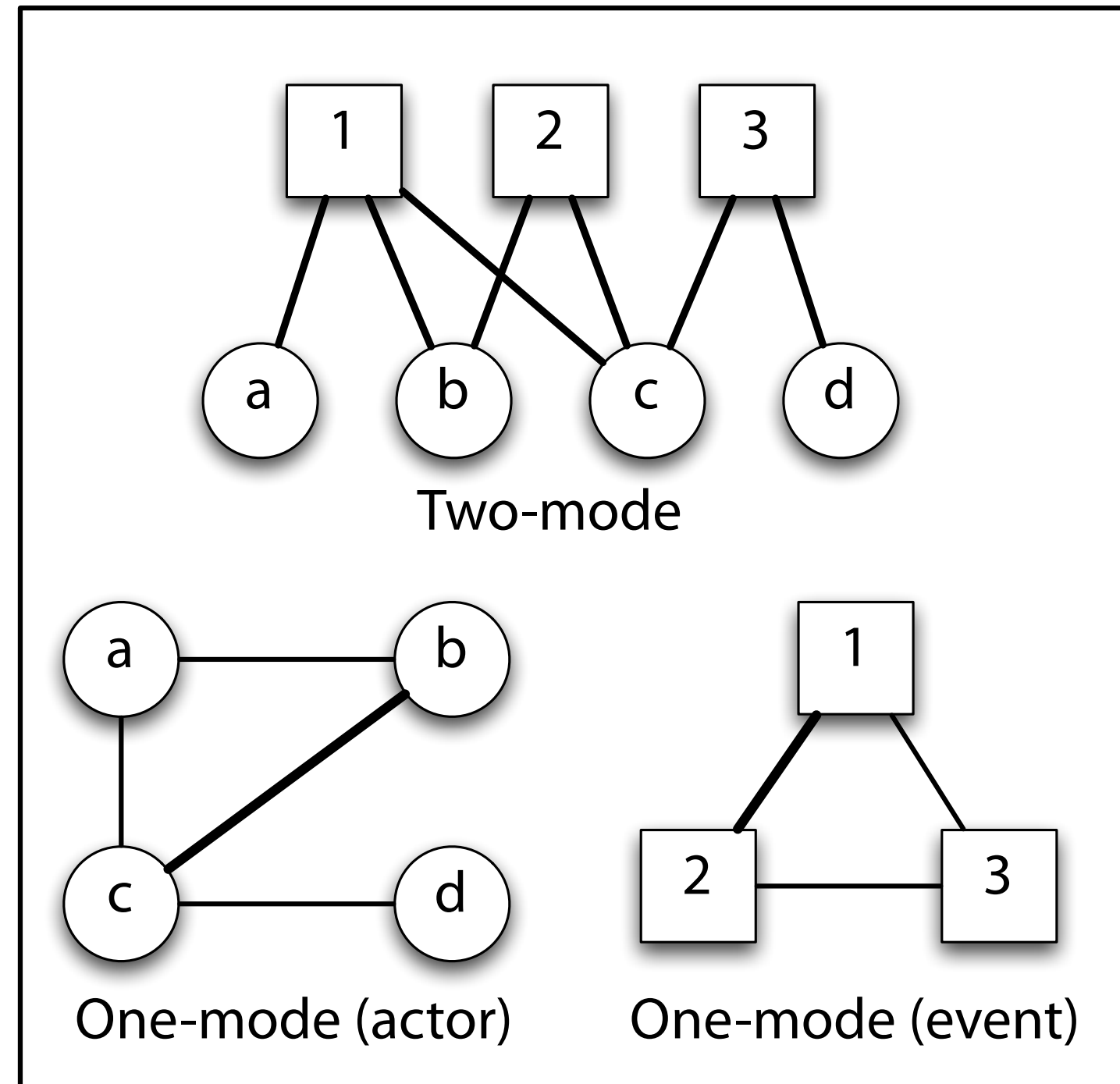


## **Communities:**

Clusters of nodes tend to group together. Where there are more nodes within a group than between a group we consider this a community.

*Community detection* is one of the most common tools in network analysis outside of looking for node position. Communities tend to emerge based on the micro level processes previously discussed.

# Networks Data: Modes



## Modes:

Networks of different types. For example, comments on a Facebook page.

## How to view the network:

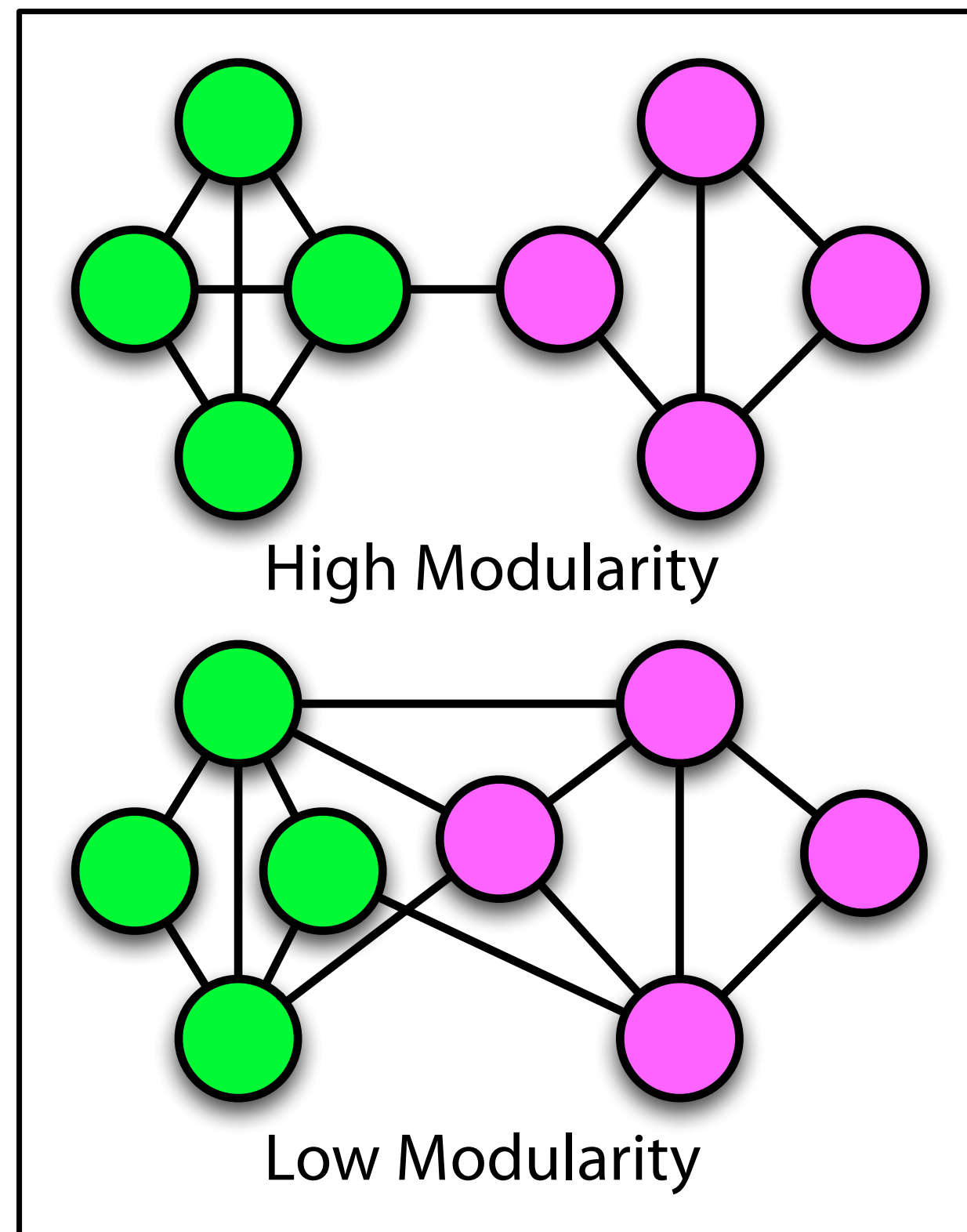
Two modes often make formatting difficult, but show both commenters and posts.

One mode projects show “co-activity”.

An edge exists in a co-activity network, if two people comment on the same post, or two posts share the same commenter



# Units of analysis: Meso



## Modularity and Communities:

When you run community detection it is trying to find a solution to maximize modularity (the measure of edges within a group to nodes between groups).

High modularity means clear groups.  
Low modularity means fuzzy groups.

A score  $> 0.3$  usually considered a good fit.

Installing Network Canvas: <https://networkcanvas.com/download>



### Interviewer

A tool for administering interviews in the field

 WINDOWS

 MACOS

 LINUX

GET IT ON  
 Google Play



### Architect

A tool for building Network Canvas interview protocols

 WINDOWS

 MACOS

 LINUX







 Network Canvas

Architect

A tool for building Network Canvas Interviews

6.4.0

Show welcome

### Welcome to Architect!

If you are new to the software, please consider watching the overview video to the left. It will explain how the software works, and introduce all the essential skills needed to build an interview protocol. We also have extensive tutorials and information on a range of topics on our documentation website, which you can visit using the link below.

Alternatively, to get started right away use the buttons below to create a new interview protocol, or open an existing one from elsewhere.

WATCH OVERVIEW VIDEO

VISIT DOCUMENTATION WEBSITE



## Resume Editing

**OSN20\_week3b\_wholeNetDe-  
mo\_redacted (schema version 7)**

/Users/work/Documents/Teaching/OSN2020/Exercises/OSN20\_wee  
(schema version 7).netcanvas

**SB22\_architect\_demo**

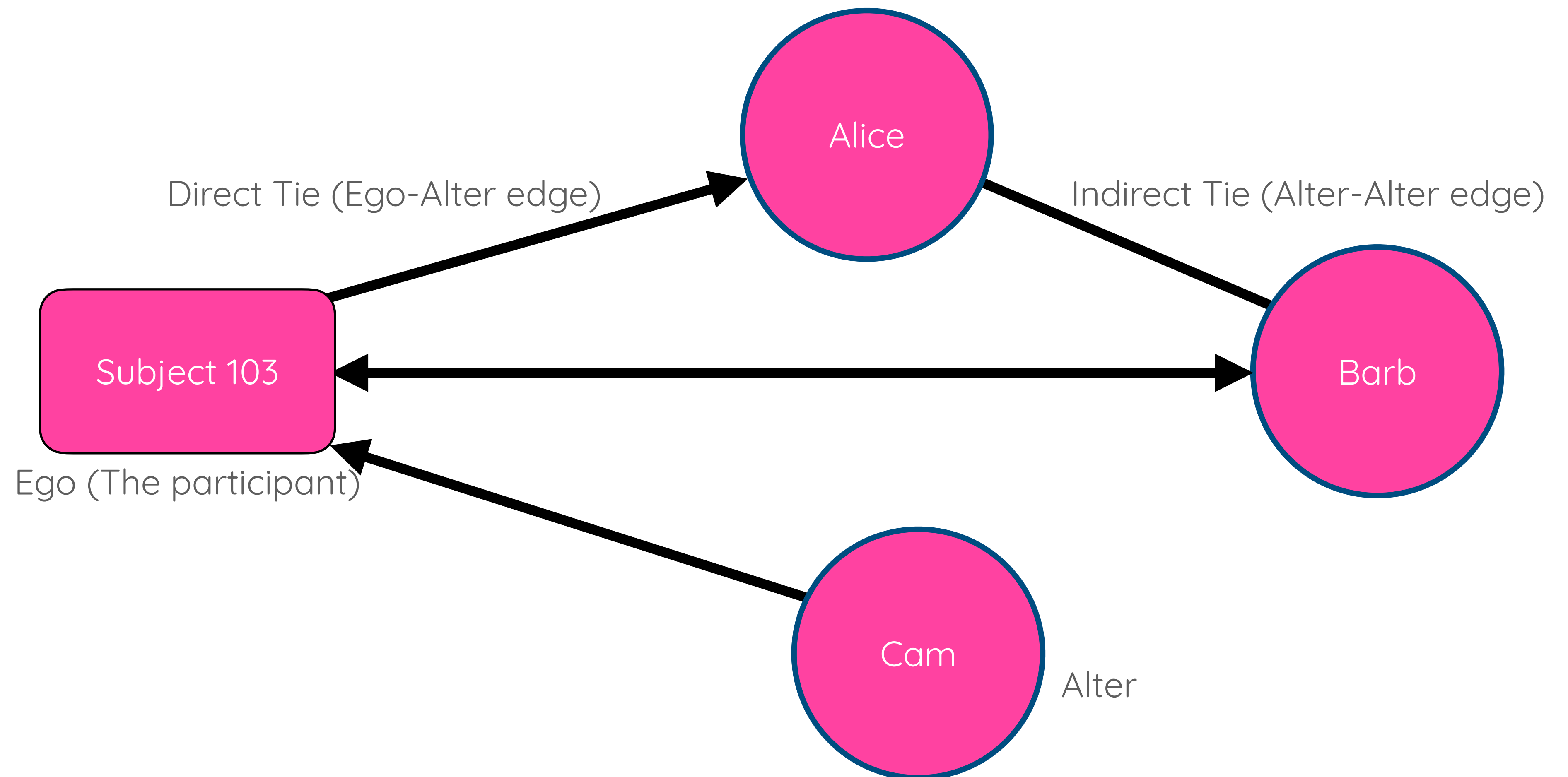
/Users/work/Documents/Teaching/Sunbelt/SB22\_architect\_demo.netcanvas

**SB22\_workshop\_protocol**

/Users/work/Documents/Teaching/Sunbelt/SB22\_workshop\_protocol.netc...

# First principles

## Network software with network objects

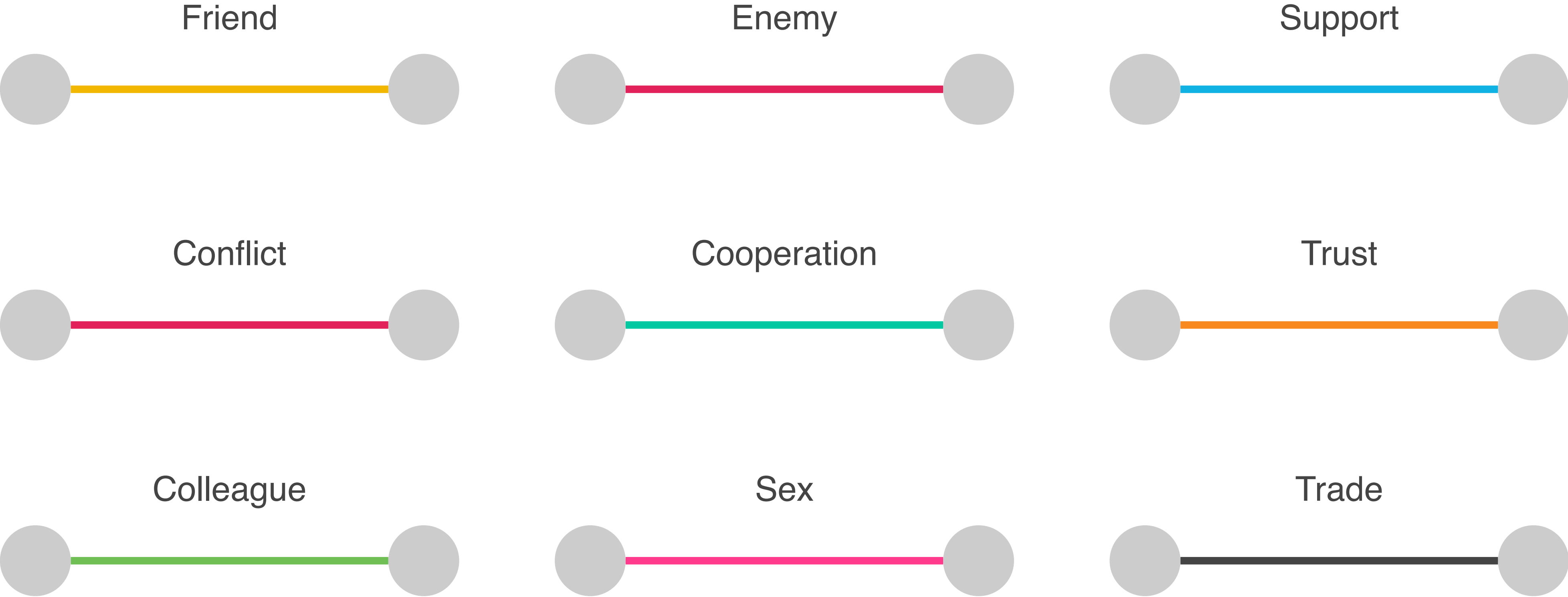


# Multiple node types (user defined)





# Multiple edge types (user defined)



An **interface** is an abstract 'type' of data collection instrument, designed with specific interaction affordances, which facilitates a core task within the network interview.

**Example:** "The categorical bin interface"

A **stage** is a specific instance of an interface, configured by you to collect the data needed in your interview. **Example:**

"My gender categorical bin stage"

A prompt is a specific question that elicits a response. Some stages have a single prompt, some can have multiple.

**Example:** "The first name generator prompt on a Name generator quick add stage"



Name Generator (using forms)

Within the **past 6 months**, who have you felt **close to**, or discussed **important personal matters** with?

Person

Person 2

Person 3

Person 4

Person 5

Name Generator (using quick add)

Within the **past 6 months**, who have you felt **close to**, or discussed **important personal matters** with?

Person

Person 2

Person 3

Person 4

Person 5

Type a name and press enter...

Small Roster Name Generator

previous interview JSON

Sort: NICKNAME Add

Filter: Filter items...

|                                           |                                             |
|-------------------------------------------|---------------------------------------------|
| <b>Eustace</b><br>Name: Nguyen<br>Age: 24 | <b>Anita</b><br>Name: Li<br>Age: 23         |
| <b>Barry</b><br>Name: Smith<br>Age: 23    | <b>Carlito</b><br>Name: Gonzalez<br>Age: 23 |
| <b>Dee</b><br>Name: Zhang<br>Age: 23      | <b>Eugene</b><br>Name: Miller<br>Age: 23    |

Large Roster Name Generator

Which classmates would you also think of as friends?

Tap an item to select it

**Eugene**  
Name: Miller  
Age: 23

**Eustace**  
Name: Nguyen  
Age: 24

**Dee**  
Name: Zhang  
Age: 23

**Anita**  
Name: Li  
Age: 23

**Carlito**  
Name: Gonzalez

2

E

Dyad Census

Are these people friends?

William

Frank

Yes

No

Tie-Strength Census

Thinking about each pair of people, please choose the option that best describes how close they are to one another.

William

Thomas

Very close

Sort of close

Not very close

Don't know each other

Ordinal Bin



Categorical Bin



Per Alter Form

James

Required likert scale

Three Two One Missing value

Does this person have this boolean attribute?

Which of these does this person like?

Politics Weather Work Current events Entertainment School Friends Money

Narrative

Per Alter Edge Form

Person Three Person Four

How often do these two people spend time together?

Very often Somewhat often Occasionally Rarely Almost never

Which category best describes the relationship between these two people?

Friends Family Co-workers Romantic Partners Other

Approximately when did these people first meet?

20th / June Clear

Ego Form

Introduction to the ego form

There are multiple ways to describe you.

What is your name?

Tell us about yourself maybe

What is your age?

ht



# How to include network objects

Ego

Alter

Direct tie (ego-  
alter)

Indirect tie  
(alter-alter)

## Select an Interface Type

🔍 Search interfaces by name or keyword...

Filter by  
capabilities:



CREATE NODES



CREATE EDGES



CAPTURE EGO DATA



CAPTURE NODE ATTRIBUTES



CAPTURE EDGE ATTRIBUTES



USE ROSTER DATA



DISPLAY MEDIA

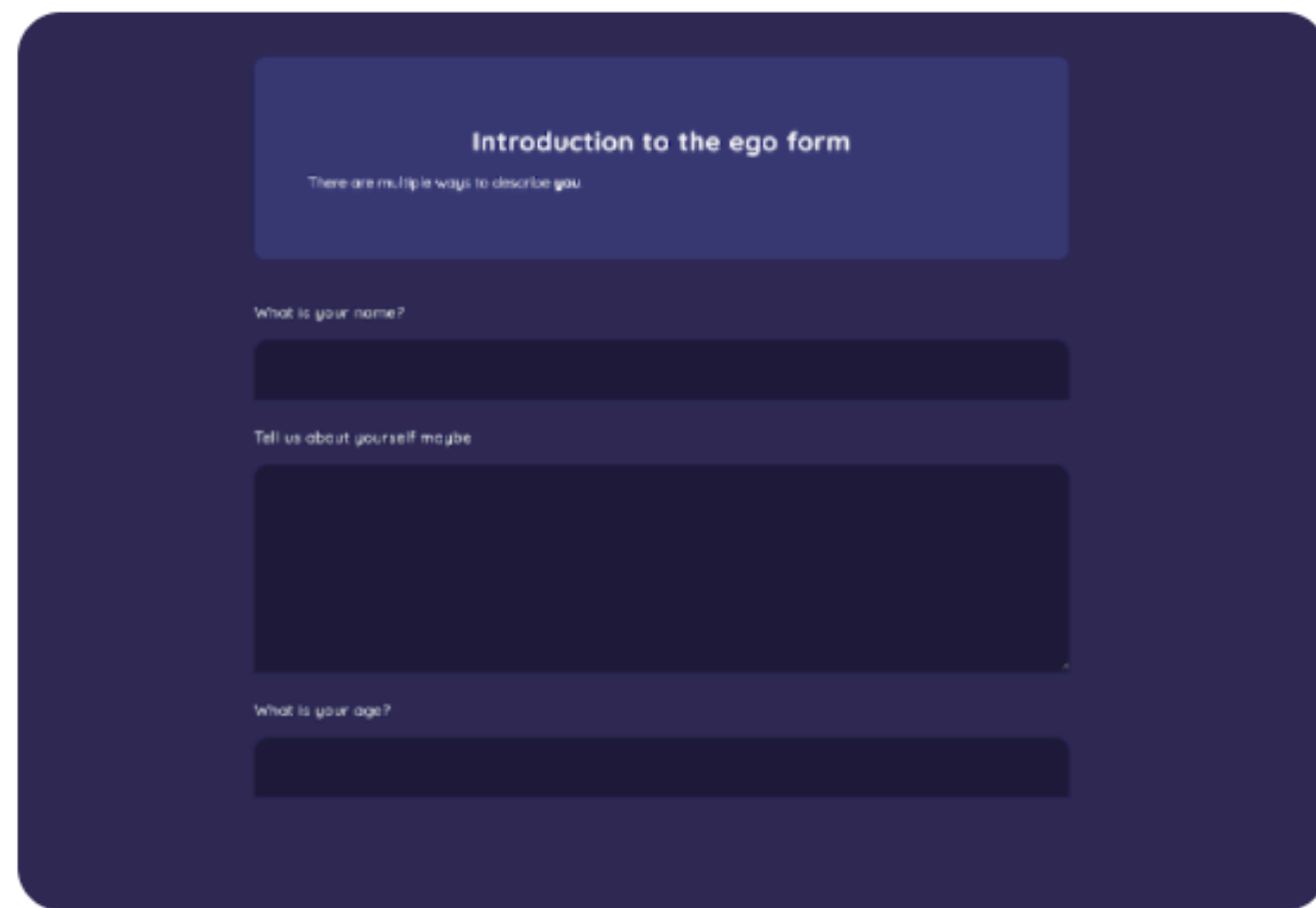


DISPLAY DATA



# Including Ego

- Generally through a participant ID (in interviewer)
- Data through an ego “form”.



Introduction to the ego form

There are multiple ways to describe you

What is your name?

Tell us about yourself maybe

What is your age?

## Ego Form

An interface that collects data on your participant (ego).



CAPTURE EGO DATA

# Including Alters



## Name Generator (quick add)

A name generator interface designed for low response-burden. Only requires a label in order to create an alter.

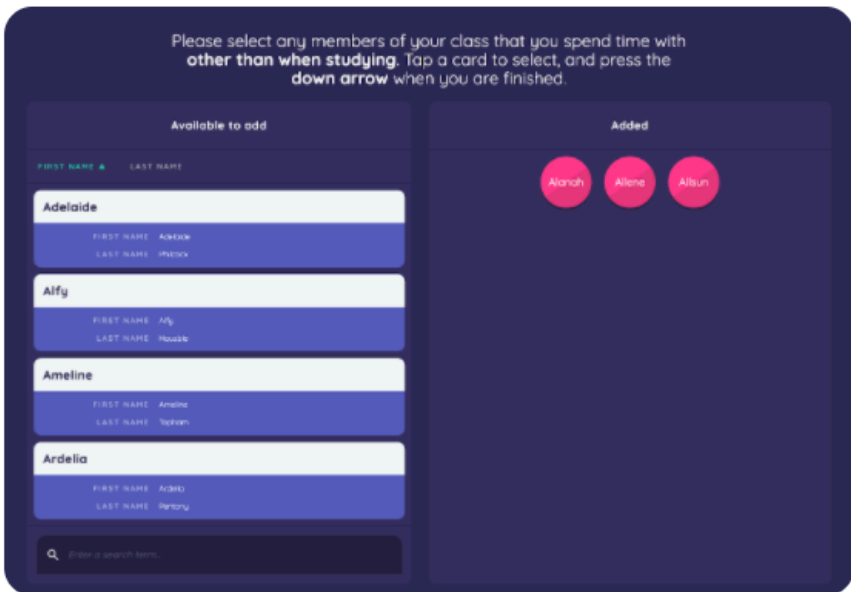
- CREATE NODES
- USE ROSTER DATA



## Name Generator (using forms)

A name generator interface which provides a form that participants complete when creating an alter.

- CREATE NODES
- CAPTURE NODE ATTRIBUTES
- USE ROSTER DATA



## Name Generator for Roster Data

A name generator specifically for roster data, allowing sorting and filtering of the roster.

- CREATE NODES
- USE ROSTER DATA

# Including Ego-Alter ties



## ASSIGN ADDITIONAL VARIABLES

This feature allows you to assign a variable and associated value to any nodes created on this prompt. You could then use this variable in your skip logic or stage filtering rules.



Select an existing variable, or select "create new variable" from the bottom of the list, and then assign a value. You can set different values for this variable for nodes created on different prompts.

### CREATE OR SELECT A VARIABLE



share\_meal



CHANGE VARIABLE



### SET VALUE OF VARIABLE TO:



True



False



# Using a side-panel

## Keeping nodes consistent

Prompt 1.

Please nominate people that you have shared a meal with in the last month

Alice

Barb

Cam

Prompt 2.

Please nominate anyone with whom you have discussed important matters. *New people can be selected using the button in the lower right, previously nominated people can be dragged over from the...*

Previously nominated

Alice

Barb

Cam

# Using a side-panel

## Keeping nodes consistent

SIDE PANELS

Use this section to configure up to two side panels on this name generator.

PANEL TITLE

The panel title will be shown above the list of nodes within the panel.

Previously nominated

DATA SOURCE

Choose where the data for this panel should come from (either the in-progress interview session ["People you have already named"], or an external network data file that you have added).

☒

 Use the network from the in-progress interview

☐

 Use a network data file

FILTER

You can optionally filter which nodes are shown on in this panel.

# Including Ego-Alter ties

## PROMPTS\*

Add one or more prompts below to frame the task for the user. You can reorder the prompts using the draggable handles on the left hand side.



Please nominate people that you have shared a meal with in the last month



Please nominate anyone with whom you have discussed important matters. *New people can be selected using the button in the lower right, previously nominated people can be dragged over from the panel on the left.*



CREATE NEW



# Including Alter- Alter ties

Filter by capabilities:

CREATE NODES

CREATE EDGES

CAPTURE EGO DATA

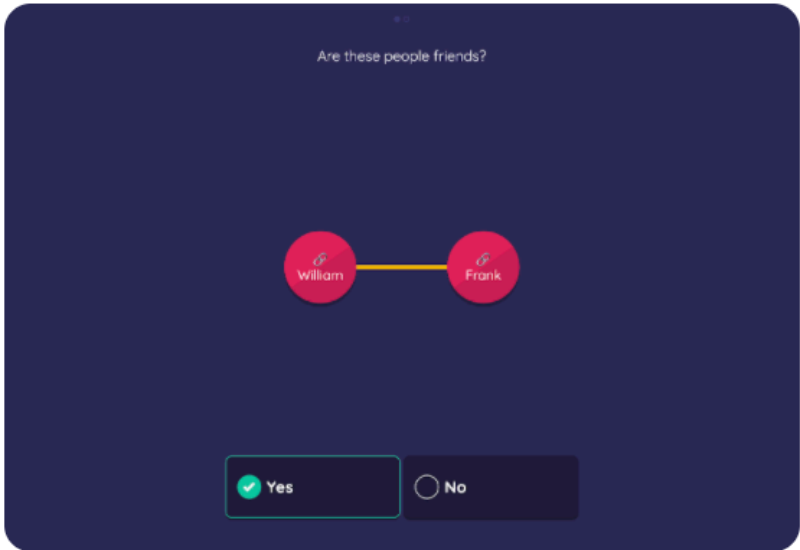
CAPTURE NODE ATTRIBUTES

CAPTURE EDGE ATTRIBUTES

USE ROSTER DATA

DISPLAY MEDIA

DISPLAY DATA



## Dyad Census

A name interpreter interface that creates edges by systematically surveying all alters in the interview network.

CREATE EDGES



## Tie-Strength Census

Combines a dyad census with an ordinal variable to simultaneously capture the strength of ties between alters.

CREATE EDGES

CAPTURE EDGE ATTRIBUTES



## Sociogram

Designed for spatially arranging alters (either manually or automatically), creating edges between them, and highlighting the presence of alter attributes.

CREATE EDGES

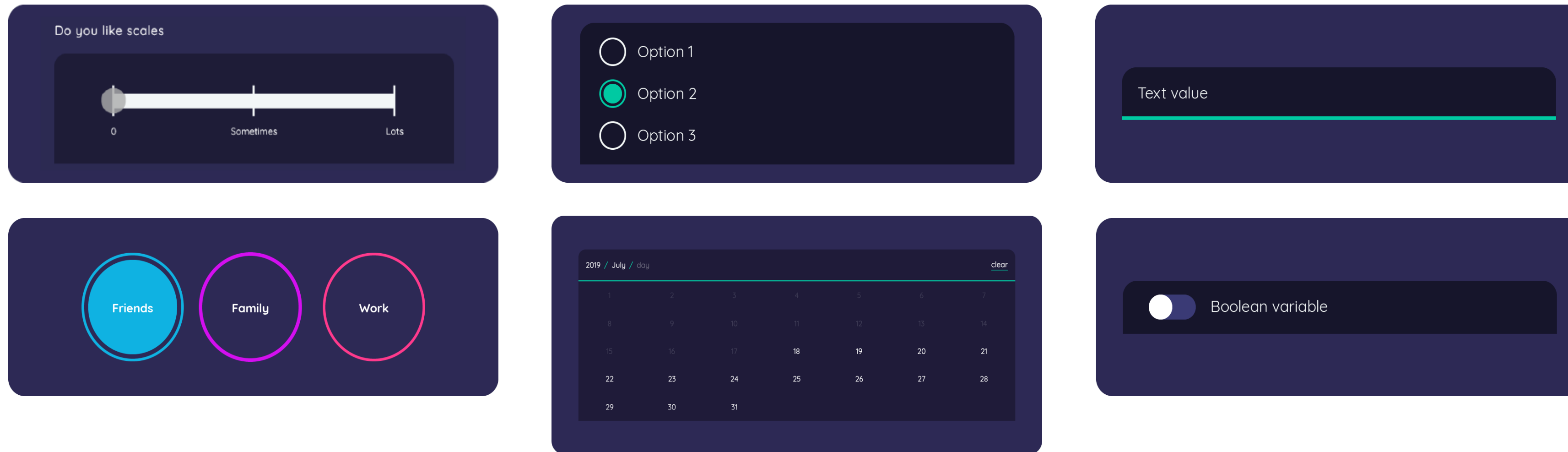
CAPTURE NODE ATTRIBUTES

DISPLAY MEDIA

# Including data about network objects

|                    |                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Text</b>        | Basic short text string,                                                                                     |
| <b>Number</b>      | Basic whole number (integer),                                                                                |
| <b>Ordinal</b>     | Ordered categorical data, where each value is described by a label and a value. Only supports single values, |
| <b>Categorical</b> | Unordered categorical data where each value is described by a label and a value. Supports multiple values.   |
| <b>Boolean</b>     | True or False values,                                                                                        |
| <b>Layout</b>      | X/Y coordinate values, normalized between 0 and 1 in each axis.                                              |
| <b>DateTime</b>    | Date variable capable of storing resolution down to the day.                                                 |
| <b>Scalar</b>      | A continuous floating point number between 0 and 1                                                           |

# Data type comes from an input control



Different input controls will have different features, like the ability for a calendar to start from 'today' or have a specific date range.

The input controls are all listed under key concepts in the documentation.



# Variables get a name, then an input type

CREATE OR SELECT A VARIABLE

? test

+ CHANGE VARIABLE

INPUT CONTROL\*

Choose an input control that should be used to

✓ -- Select an input control --

-- Number Types --

Number Input

-- Scalar Types --

VisualAnalogScale

-- Date Types --

DatePicker

RelativeDatePicker

-- Text Types --

Text Input

Text Area

-- Boolean Types --

BooleanChoice

Toggle

-- Ordinal Types --

Radio Group

LikertScale

-- Categorical Types --

Checkbox Group

Toggle Button Group

CREATE OR SELECT A VARIABLE

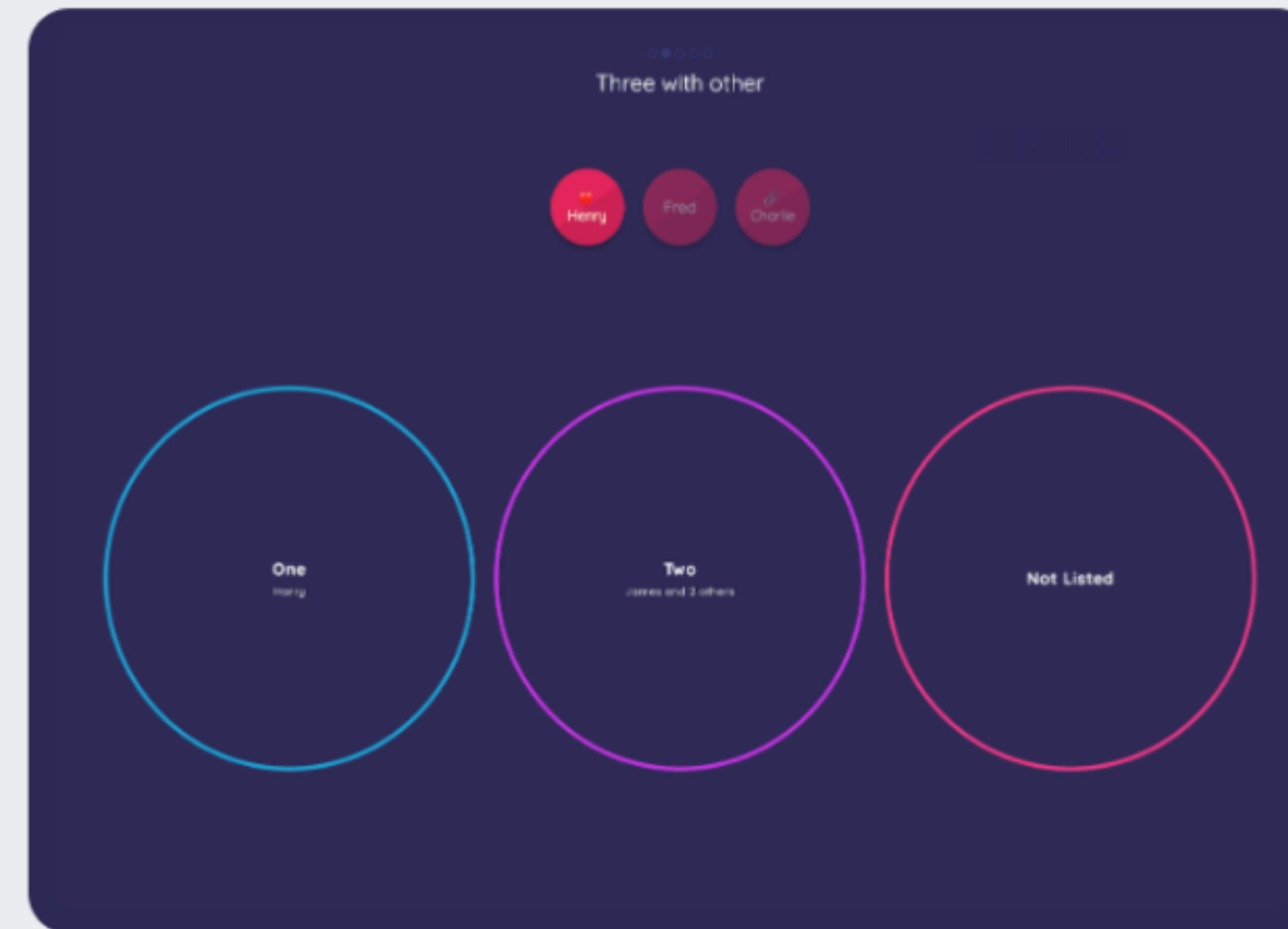
test

+ CHANGE VARIABLE

### Ordinal Bin



### Categorical Bin



### Per Alter Form

James

Required likert scale

☐ Three

☐ Two

☐ One

☐ Missing Value

Does this person have this boolean attribute?

☐

Which of these does this person like?

Politics

Weather

Work

Current events

Entertainment

School

Friends

Money

Submit

## Narrative

## Per Alter Edge Form

Person Three

Person Four

How often do these two people spend time together?

Very Often

Somewhat Often

Occasionally

Rarely

Almost Never

Which category best describes the relationship between these two people?

Friends

Family

Co-workers

Romantic Partners

Other

Approximately when did these people first meet?

2011 / June

clear

1 of 4

## Ego Form

## Introduction to the ego form

There are multiple ways to describe you.

What is your name?

Tell us about yourself maybe

What is your age?



ht

# Assigning data in steps

## A general approach

- Select a Stage based on whether you are collecting Ego, Node, or Edge data;
- Include the type of ego, node, or edge;
- Create a variable for that data;
- Select an input control;
- Network Canvas will then assign the variable a type based on that control.



# Selecting the data collection “approach”

## Per network approaches versus per object approaches

Per object:

- Ego
- Alter
- Edge

Introduction to the ego form

There are multiple steps to describe ego.

What is your name?

Tell us about yourself maybe

What is your age?

Required alert score

Does this person have this boolean attribute?

Which of these does this person like?

How often do these two people spend time together?

Which category best describes the relationship between these two people?

Approximately when did these people first meet?

Per network:

- Bins
- Sociograms

Three with other

One

Two

Not Listed

Contact frequency, ordinal variable for close ties only

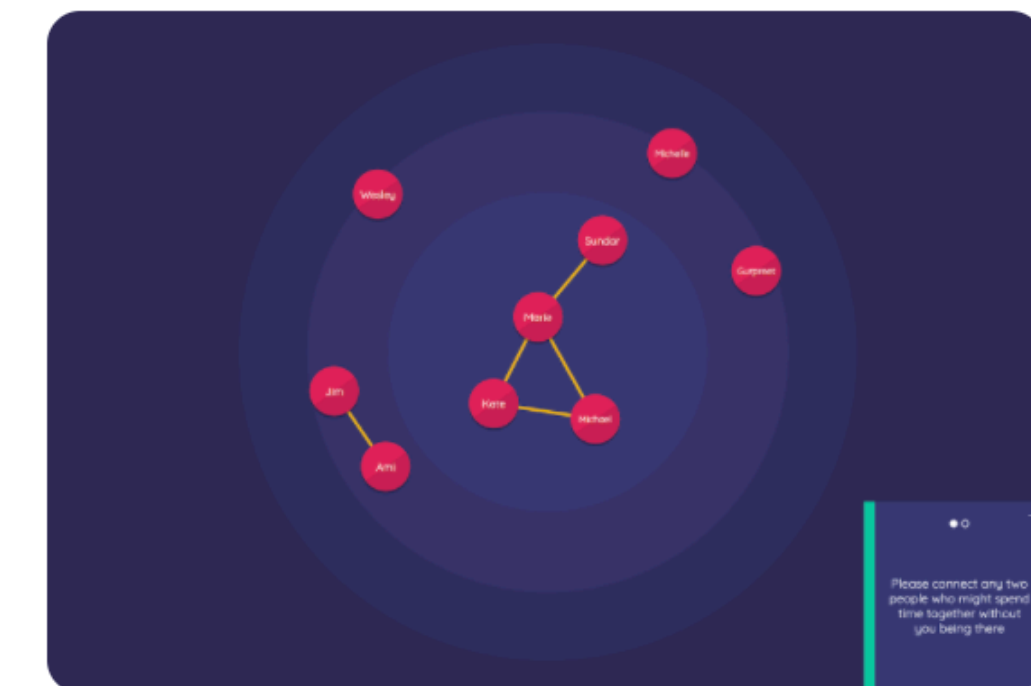
Last 24 hours

Last week

Last six months

Last two years

Over two years ago



# Considering the following questions

Maybe I'm just hungry

“How often do you go for a meal with this person”,

“Select some cuisines that you would eat with this person”

“To the best of your knowledge, is this person a vegetarian or vegan?”

- What are the data types that we are expecting?
- What input controls would work for which data types?
- Should these be per network or per alter?

# Network Canvas Gotchas

Some things about style learned from experience

- Form length and missing data
- Duplicate nodes
- Boring interviews



# Dealing with form length

- Example from SB22\_architect\_demo.netcanvas
- Warnings and mandatory questions
- You can also use validation options to help this.

FIELDS\*

Add one or more fields to your form to collect attributes about each node the participant creates. Use the drag handle on the left of each prompt adjust its order.

≡

Which colours do you like to wear

Categorical variable using CheckboxGroup input control

×

≡

What is your favourite colour?

Text variable using Text input control

×

+

CREATE NEW

VALIDATION

Add one or more validation rules to this form field.

required

▼

×

+

ADD NEW

# Duplicate nodes

- Side panels which can be used for these things to remind people of previously mentioned alters.
- It's important to think of node types as constitute separate alter pools that you draw upon in different stages and prompts.
- Make sure people do not nominate a node twice or three times.
- You can also set minimum and maximum number of alters.

# Boring interviews

## Try to keep it fresh

- Use the fact that variables are passed around in order to break up the interview.
- Remember that a layout variable is not stage-specific so you can alternate with sociograms and other questions.
- Think about the ways in which you are asking the participant. Asking about disparate questions of an alter or a cohesive questions? Think per network forms or per alters.
- Break up forms, don't just use validation.



# Ideas for whole network work

## Using Rosters

- See the demonstration using `SB22_whole_network_demo.netcanvas`
- You can upload a roster as a CSV file. Be sure to have a column with name, preferably in the first column.
- Let's open in Architect to see the previous screen.

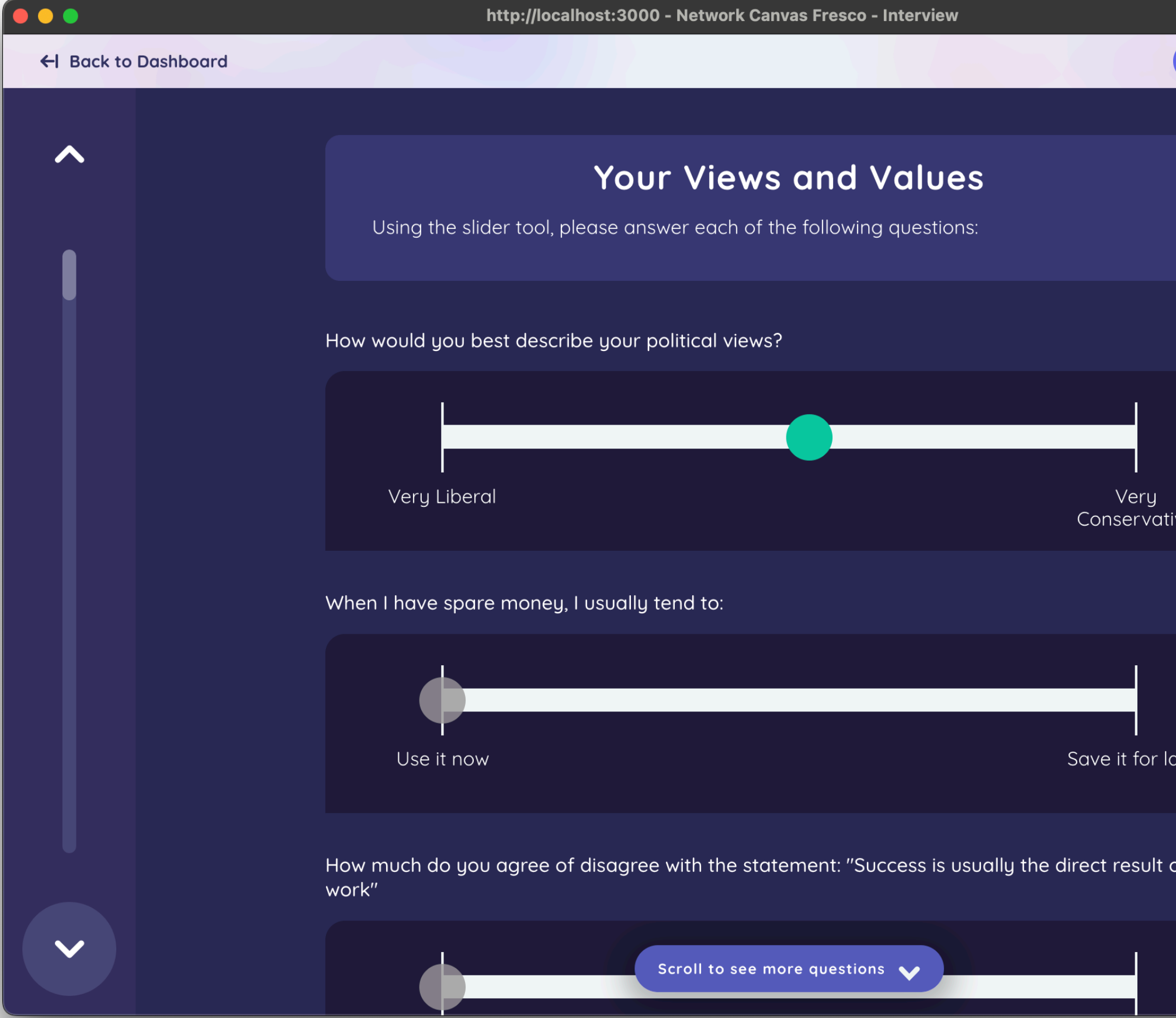
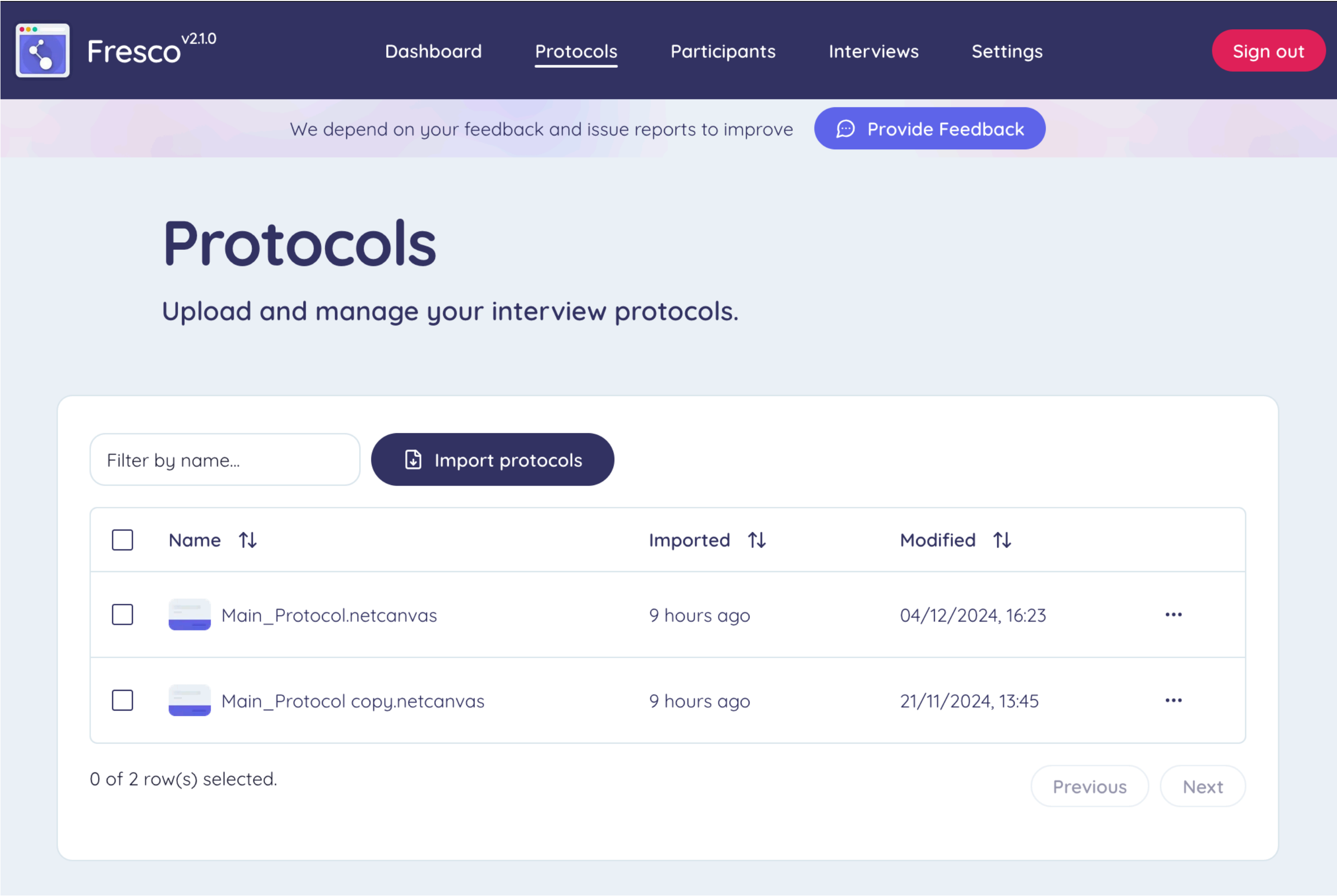
Please select people that you knew before coming to class

| Available to add                                   | Added |
|----------------------------------------------------|-------|
| <div>a</div> <div>FIRST NAME a<br/>SURNAME a</div> |       |
| <div>b</div> <div>FIRST NAME b<br/>SURNAME b</div> |       |
| <div>c</div> <div>FIRST NAME c<br/>SURNAME c</div> |       |
| <div>d</div> <div>FIRST NAME d<br/>SURNAME d</div> |       |
| <div>e</div> <div>FIRST NAME e</div>               |       |

🔍 Enter a search term...

# New this year and upcoming

## Introducing Fresco!



# Notes about Fresco

**It is not as easy to install or configure**

- Fresco is a web portal that is built with similar components as Interviewer but needs a server to run on.
- The documentation provides an example way to do this using a third party server, and the readme provides a way to do local hosting using software called Docker.
  - This involves a little familiarity with GitHub (and ideally some JavaScript).
- We understand this. Our medium term goal (i.e. within two years) is a platform “Network Canvas Studio” where you can host network canvas surveys.



# Now - Into the Protocols!

- We have three demonstration protocols to consider.
- I will walk through the first one with everyone to also show how to install a protocol.
  - Importantly, I will be talking through these in ARCHITECT while also doing these in INTERVIEWER.

# Your turn!

## Thank you so much

- What sort of tasks did you want to accomplish with Network Canvas?
- What challenges have you had?



Network Canvas

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